

Task 19 — Analytical Data Pipeline Packing

Data Analyst · Phase 1 Industry Immersion · Study Guide

DATA ANALYST

PlaceMux · Phase 1 · Study Guide

DIFFICULTY

Mastery · Expert

Level 19 of 20 · ramp 90%

1 · What this is & why it matters

Package the analytical workflow into a reusable, parameterised pipeline that runs end-to-end on fresh data. One-off notebooks don't scale. A packaged pipeline makes analysis repeatable, schedulable and trustworthy.

2 · Learning objectives

By the end of this guide and task you will be able to:

- Deliver: A parameterised, logged, re-runnable analytical pipeline with a one-command run.
- Explain the key concepts behind the task in your own words.
- Choose between alternative ways to execute it, and justify your choice.
- Avoid the common pitfalls and know how your work is scored.

3 · Prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.

4 · Key concepts (understand these first)

Pipeline

ordered, automated steps from raw data to outputs.

Parameterisation

run the same pipeline on different inputs/dates.

Idempotence

re-running produces the same result, no duplicates.

Modularity

each step is a testable function.

Logging & checkpoints

you can see what ran and resume on failure.

5 · Recommended stack & tools

- Python modules
- a runner (Make/CLI/Prefect/Airflow-lite)
- logging

6 · The build pipeline (follow in order)

This is the flow from nothing to demoable. Each step builds on the last; the final steps verify and prove the work.

1. Refactor notebook steps into clean, named functions.
2. Chain them into a single pipeline with clear inputs/outputs.

3. Parameterise dates/sources so it runs on new data.
4. Add logging and basic data-quality checks between stages.
5. Make outputs idempotent and versioned.
6. Run the whole pipeline on a fresh slice to prove it.

7 · Other ways to execute (alternative approaches)

There is rarely one right path. Consider these alternatives and pick deliberately for your context:

- Lightweight script pipeline vs a full orchestrator (Airflow/Prefect).
- Pandas pipeline vs pushing steps into SQL.

8 · Brainstorming & deeper thinking

Sit with these questions — they separate someone who completed the task from someone who understood it:

- What happens if step 3 fails halfway -- does it corrupt outputs?
- Which step is most likely to break on new data, and how is it guarded?
- How will someone else run this without you?

9 · Definition of Done

Done when all are true and demoable

- A parameterised, logged, re-runnable analytical pipeline with a one-command run.
- Demonstrable live on real data.

10 · How your work is evaluated (out of 100)

Scoring parameter	Marks
Core deliverable — A parameterised, logged, re-runnable analytical pipeline with a one-command run.	50
Real-data quality & correctness (real inputs at realistic scale, not a toy/happy-path)	20
Live verification & evidence (demonstrated live; real output, not claims)	15
Dependency, failure & edge-case handling (errors handled; hand-offs honoured)	15
TOTAL	100

11 · Pitfalls to avoid

Worry if any of these are true of your work

- Leaving it as a fragile notebook.
- No checks between stages.
- Non-idempotent steps that duplicate outputs on re-run.

12 · Where this fits & go deeper

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.

Go deeper (optional further study):

- Statistics for analysts (distributions, tests, intervals)
- SQL window functions and query optimisation
- Storytelling with data and dashboard design

- Time-series and segmentation techniques